

Experiment 11

Question: Experiment to understand pandas library use cases in Data science.

Aim

To understand the use cases of the pandas library for data manipulation and analysis in data science.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def exp_11():
    data = {"Name": ["Asha", "Ravi", "Meena", "Kumar", "Divya", "Arun"],
            "Dept": ["IT", "HR", "IT", "Sales", "HR", "IT"],
            "Salary": [55000, 42000, 61000, 38000, 45000, 70000],
            "Experience": [3, 5, 6, 2, 4, 9]}
    df = pd.DataFrame(data)          # 1. create DataFrame
    print("DataFrame:\n", df)
    df.to_csv("employees.csv", index=False)  # 2. write / read CSV
    df = pd.read_csv("employees.csv")

    print("\nShape:", df.shape)          # 3. inspect
    print(df.dtypes)
    print(df.describe())

    print("\nSelect column 'Name':", list(df["Name"]))          # 4. selection
    print("loc row 2 : \n", df.loc[2])
    print("iloc rows 0-1: \n", df.iloc[0:2, :3])

    print("\nSalary > 50000: \n", df[df["Salary"] > 50000])  # 5. filtering
    print("\nSorted by Salary (desc): \n", df.sort_values("Salary", ascending=False))

    df["Bonus"] = df["Salary"] * 0.10          # 6. new column
    df["Level"] = df["Experience"].apply(lambda x: "Senior" if x >= 5 else "Junior")
    print("\nWith new columns: \n", df)

    print("\nGroupby Dept -> mean Salary: \n", df.groupby("Dept")["Salary"].mean())
    print("\nMultiple aggregations: \n",
          df.groupby("Dept").agg(Count=("Name", "count"),
                                AvgSalary=("Salary", "mean"),
                                MaxExp=("Experience", "max")))
    print("\nvalue_counts of Level: \n", df["Level"].value_counts())

    dept = pd.DataFrame({"Dept": ["IT", "HR", "Sales"],
                          "Head": ["Suresh", "Priya", "Anil"]})
    print("\nMerge with department heads: \n", df.merge(dept, on="Dept")["Name", "Dept", "Head"])
    print("\nPivot table: \n", df.pivot_table(values="Salary", index="Dept",
                                              columns="Level", aggfunc="mean"))
```

exp_11()

Output

```
DataFrame:
   Name  Dept  Salary  Experience
0  Asha   IT   55000         3
1  Ravi   HR   42000         5
2  Meena  IT   61000         6
3  Kumar Sales  38000         2
4  Divya  HR   45000         4
5  Arun   IT   70000         9
```

Shape: (6, 4)

```
Name      str
Dept      str
Salary    int64
Experience int64
dtype: object
```

```
      Salary  Experience
count  6.000000  6.000000
```

```

mean    51833.333333    4.833333
std     12319.361455    2.483277
min     38000.000000    2.000000
25%     42750.000000    3.250000
50%     50000.000000    4.500000
75%     59500.000000    5.750000
max     70000.000000    9.000000

```

```
Select column 'Name': ['Asha', 'Ravi', 'Meena', 'Kumar', 'Divya', 'Arun']
```

```

loc row 2 :
  Name      Meena
Dept        IT
Salary      61000
Experience   6
Name: 2, dtype: object
iloc rows 0-1:
  Name Dept Salary
0  Asha  IT  55000
1  Ravi  HR  42000

```

```

Salary > 50000:
  Name Dept Salary Experience
0  Asha  IT  55000          3
2  Meena IT  61000          6
5  Arun  IT  70000          9

```

```

Sorted by Salary (desc):
  Name Dept Salary Experience
5  Arun  IT  70000          9
2  Meena IT  61000          6
0  Asha  IT  55000          3
4  Divya HR  45000          4
1  Ravi  HR  42000          5
3  Kumar Sales 38000          2

```

```

With new columns:
  Name Dept Salary Experience Bonus Level
0  Asha  IT  55000          3  5500.0 Junior
1  Ravi  HR  42000          5  4200.0 Senior
2  Meena IT  61000          6  6100.0 Senior
3  Kumar Sales 38000          2  3800.0 Junior
4  Divya HR  45000          4  4500.0 Junior
5  Arun  IT  70000          9  7000.0 Senior

```

```

Groupby Dept -> mean Salary:
Dept
HR      43500.0
IT      62000.0
Sales   38000.0
Name: Salary, dtype: float64

```

```

Multiple aggregations:
  Count AvgSalary MaxExp
Dept
HR      2    43500.0      5
IT      3    62000.0      9
Sales   1    38000.0      2

```

```

value_counts of Level:
Level
Junior    3
Senior    3
Name: count, dtype: int64

```

```

Merge with department heads:
  Name Dept Head
0  Asha  IT  Suresh
1  Ravi  HR  Priya
2  Meena IT  Suresh
3  Kumar Sales Anil
4  Divya HR  Priya
5  Arun  IT  Suresh

```

```

Pivot table:
Level Junior Senior
Dept
HR      45000.0  42000.0
IT      55000.0  65500.0
Sales   38000.0    NaN

```